When Mean-Residual Distillation Fails: Frozen Decision-Level Correction for AFDM Joint Receivers

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***Abstract*—AFDM receivers can have nontrivial oracle headroom in doubly selective channels, but exploiting that headroom with learned detector-side residuals is harder than a standard regression view suggests. We revisit this problem in a corrected AFDM joint-receiver evaluation and show that the original mean-residual distillation story collapses once the evaluation bug is removed: the useful headroom is concentrated in sparse decision-boundary crossings rather than smooth mean-level corrections. Guided by this diagnosis, we replace residual prediction with a lightweight frozen bit-level corrector that uses only observable receiver trajectory features. The corrector is trained once on one dedicated cache and then frozen for untouched test caches and shifted regimes. On three untouched nominal caches, it improves the mean BER from 6.3553×10^{-2} to 6.2493×10^{-2}, giving an average gain of 1.0590×10^{-3}. The same frozen model remains positive under higher Doppler severity, an integer-Doppler structural shift, and lower pilot power, with paired-bootstrap confidence intervals above zero in the positive regimes. The method is not merely an LLR thresholding trick: an LLR-only variant retains only a small fraction of the full gain. We further show that the bit-level gains translate into relative goodput improvements of 3.62%–5.72% for 64-bit packets and 4.97%–11.33% for 256-bit packets. At the same time, a severe path-count shift remains a clear failure boundary. These results support a scope-bounded diagnostic-plus-method view of AFDM post-hoc receiver correction.**

***Index Terms*—AFDM, joint receiver, model-driven receiver, decision-level correction, regime shift, packet success, goodput**

I. Introduction

Affine frequency division multiplexing (AFDM) has emerged as a promising waveform for doubly dispersive wireless channels because it combines chirp-domain structure with favorable robustness to delay and Doppler effects [1]–[3]. As the AFDM literature matures, recent work has expanded from waveform fundamentals to channel estimation, joint detection, and more structured receiver designs [4]. In parallel, learned and model-driven receivers have shown that carefully chosen inductive structure can improve detection without discarding domain knowledge [5]–[7]. A natural question is whether a strong AFDM joint receiver can be improved further through a lightweight detector-side learning module.

Our project started from exactly that hypothesis. The initial idea was to distill oracle detector corrections as smooth mean-level residuals. That hypothesis looked plausible because oracle replay showed nontrivial headroom. However, once we corrected the evaluation bug in the AFDM pipeline, the headline gain disappeared: the base learned receiver, its variance-recalibrated variant, and the cleaned mean-residual distillation model showed essentially no meaningful BER separation on the same test protocol. The resulting lesson is not that detector-side adaptation is useless, but that the original target abstraction was wrong.

The central observation of this paper is that the useful headroom in the corrected AFDM setting is concentrated in sparse decision-boundary crossings rather than broad mean-level changes. This matters because small regression loss improvements do not necessarily change the final hard decisions. Guided by this diagnosis, we replace residual regression with a frozen decision-level corrector that acts directly on observable receiver trajectory features. The final method is intentionally small: a train-once multilayer perceptron that takes only observable per-bit features, runs after the base receiver, and is evaluated without retraining on untouched caches and shifted regimes.

Our contributions are:

- We show that detector-side mean-residual correction is not decision-effective in the corrected AFDM joint-receiver evaluation, even though oracle-beneficial flips exist.
- We introduce a lightweight frozen direct-bit corrector that uses only observable trajectory features and improves the mean BER by 1.0590×10^{-3} on untouched nominal caches.
- We show that the same frozen model transfers across higher Doppler severity, an integer-Doppler structural shift, and lower pilot power, while keeping a severe path-count shift as an explicit failure boundary.
- We translate the bit-level gains into packet-level utility, obtaining relative goodput gains up to 11.33% under positive regimes.

The rest of the paper follows the same diagnostic-to-method logic. Section II positions the work relative to AFDM and learned receiver literature. Section III explains the corrected evaluation and why the problem must be treated as a decision-level correction task. Section IV presents the frozen corrector and its train-once/test-on-many protocol. Section V reports nominal, transfer, ablation, and packet-level results. Section VI summarizes the scope boundary and implications.

II. Related Work

a) AFDM waveform design and AFDM receiver algorithms.: AFDM was introduced as a waveform for highly doubly dispersive channels with a chirp-domain representation that differs from both OFDM and OTFS [1]. Subsequent studies have characterized AFDM behavior on sparse time-varying channels and compared it with nearby waveform families in integrated sensing and communications settings [2], [3]. More recent receiver work has focused on AFDM-specific channel

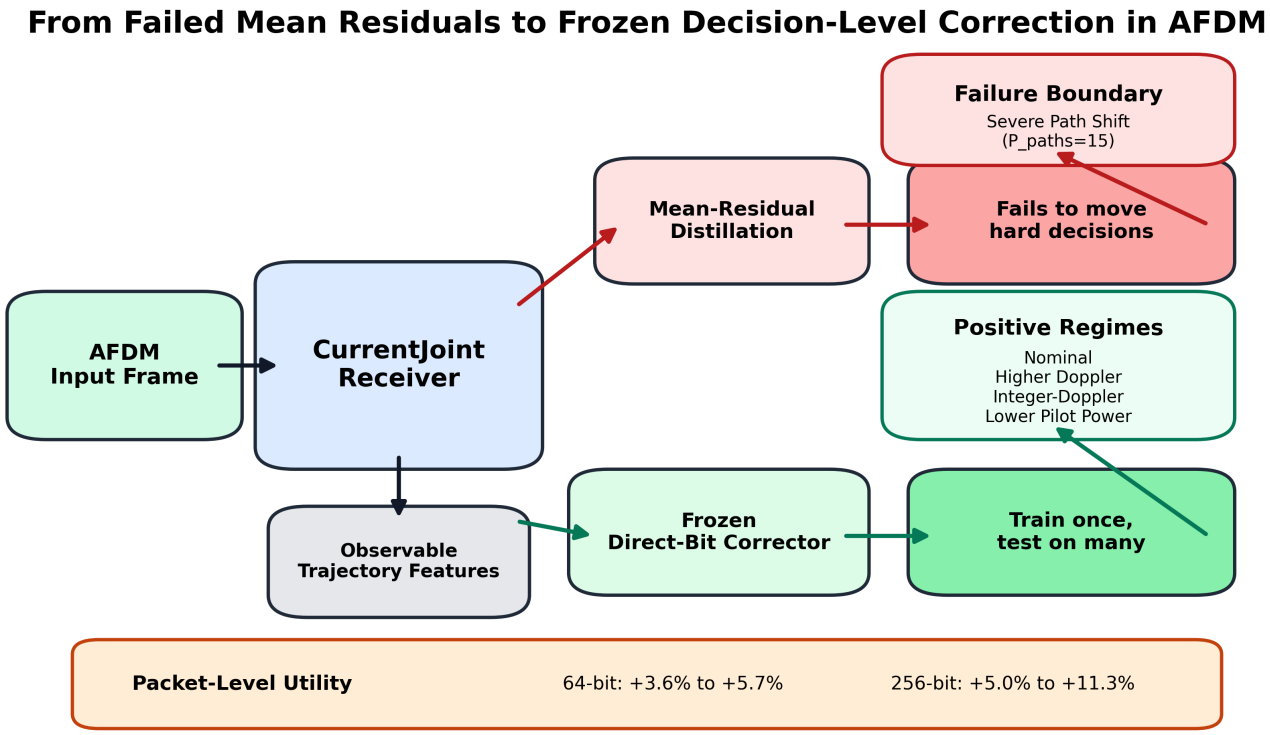


Fig. 1: Diagnostic-plus-method overview. In the corrected AFDM evaluation, useful headroom is concentrated in sparse decision-boundary crossings rather than smooth mean-level residual shifts. This motivates a frozen observable bit-level corrector, which improves BER across untouched and shifted regimes while retaining an explicit failure boundary under severe path-count shift.

estimation and detection under delay-Doppler interference, especially in MIMO or pilot-aided settings [4]. Our paper is different in emphasis: we do not propose a new AFDM waveform or a new large receiver architecture. Instead, we study how to correct the output of a fixed AFDM joint receiver after a diagnostic analysis reveals that the original residual-level learning target is misaligned with decision-level utility.

b) Model-driven and learned wireless receivers.: Hybrid and model-driven receivers have become a standard strategy in wireless communications because they preserve signal-structure priors while exploiting trainable components [5]. Representative examples include deep soft interference cancellation and decision-feedback-aware receiver learning, both of which use domain structure to constrain the learning problem [6], [7]. Our work shares the model-driven philosophy but differs in two ways. First, we do not redesign the internal iterative algorithm of the base AFDM receiver. Second, our main result is not a larger or more adaptive learned architecture, but a small frozen corrector whose form is justified by a failure analysis.

c) Post-detection refinement and decision feedback.: Several learned receivers refine symbol decisions using confidence, interference structure, or decision feedback [6], [7]. Those works show that decision-level information can be more actionable than raw regression targets when the receiver dynamics are strongly structured. Our findings are aligned with that intuition, but the route is different: we first show that mean-residual detector distillation fails under corrected AFDM evaluation, and only then move to a post-hoc decision-level corrector. The contribution is therefore not just a new corrector, but a stronger claim-evidence mapping about *why* that corrector is the right abstraction in this AFDM setting.

d) Positioning of this paper.: The closest high-level categories to our work are AFDM-specific receiver design and model-driven learned detection. Yet the final paper is neither a pure receiver redesign paper nor a pure negative-results note. The distinctive feature of our story is the transition from a failed residual-level learning route to a frozen train-once/test-on-many decision-level corrector, together with explicit transfer tests, feature ablations, packet-level utility, and a visible failure boundary.

III. Problem Setting and Diagnostic Motivation

We start from a fixed AFDM joint receiver, denoted by CurrentJoint, that produces final bit log-likelihood ratios together with internal iterative statistics. Let $\ell^{(t)}$, $v_x^{(t)}$, and $\sigma^{(t)}$ denote the per-bit detector LLRs, the variance proxy, and the effective noise statistic at outer iteration t. The original research objective was to learn a detector-side residual that would move the final output toward an oracle-improved detector state.

The corrected evaluation changed that picture. Once the pilot-format bug was removed, CurrentJoint, vx_recal_v1, and the cleaned mean-residual distillation model became almost indistinguishable on the same test protocol. In other words, the earlier positive story was an evaluation artifact, not a robust detector improvement. This forced a more careful question: if

oracle headroom exists, what kind of intervention can actually reach it?

The answer came from a decision audit. On a corrected held-out cache, the base receiver BER was 6.0352×10^{-2}, the teacher disagreement rate was 5.2051×10^{-2}, and the oracle-beneficial flip rate was 3.1543×10^{-2}. These numbers imply that useful headroom exists, but it is concentrated in a relatively sparse set of bits near the decision boundary. A smooth mean-residual correction can reduce a regression loss while still failing to change the final hard decisions.

This motivates a per-bit observable feature representation instead of a raw residual target. For each decoded bit, we form

$$\mathbf{z} = \begin{bmatrix} \ell^{(T)},\ |\ell^{(T)}|,\ \ell^{(0)},\ |\ell^{(0)}|,\ \ell^{(T)} - \ell^{(0)},\ |\ell^{(T)} - \ell^{(0)}|, \\ v_x^{(T)},\ v_x^{(0)},\ v_x^{(T)} - v_x^{(0)},\ \sigma^{(T)},\ \sigma^{(0)},\ \mathrm{SNR} \end{bmatrix} \in \mathbb{R}^{12}. \tag{1}$$

where T is the final iteration. These quantities are observable from the base receiver and do not require teacher trajectories at test time.

The resulting problem is therefore not "predict the oracle residual" but rather "predict whether the current bit decision should be corrected from observable trajectory evidence." This difference in abstraction is the key technical shift of the paper. It explains why the final method is post-hoc, frozen, and deliberately simple: the goal is to isolate a decision-level intervention that is scientifically interpretable and deployable within the current AFDM setting.

IV. Frozen Decision-Level Corrector

The final method, denoted by FROZENBITCORRECTOR, is a lightweight multilayer perceptron that maps the per-bit observable feature vector $\mathbf{z}$ to a binary correction decision. The model uses two hidden layers of sizes 32 and 16 with ReLU nonlinearities, followed by a single logit output. Training uses a binary cross-entropy objective with logits. The target is the ground-truth bit label, and the correction is applied after the base receiver has produced its final bit decisions.

Two choices are central to the method design.

a) Decision-level target.: The network predicts bit labels, not detector residuals. This is a direct consequence of the diagnostic evidence in Section III: the useful headroom is sparse and decision-level, so optimizing a mean residual is the wrong surrogate. The model is therefore trained to recover the correct final bit decision from observable receiver-state features.

b) Frozen train-once/test-on-many protocol.: The model is trained once on a dedicated cache and then frozen. No test-cache tuning, teacher replay, or split-specific adaptation is allowed. This protocol avoids the ambiguity of split-specific probes and turns the corrector into a genuine post-hoc method rather than an evaluation-specific fit. In the implementation used in this paper, the training cache is independent of all reported test caches, and the same frozen checkpoint is applied to nominal and shifted regimes.

The feature design also reflects the diagnostic constraint. The input is limited to observable statistics from the base receiver: initial and final LLRs, their absolute values and differences, initial and final variance proxies, initial and final effective noise statistics, and the frame SNR. This makes the method stronger than a pure confidence threshold, but still much simpler than a redesigned iterative detector.

At inference time, the pipeline is straightforward: run CURRENTJOINTonce, build the 12-dimensional per-bit feature tensor, evaluate FROZENBITCORRECTOR, and flip the output bit whenever the predicted probability exceeds 0.5. Because the corrector is frozen and small, its role is easy to interpret: it is a post-hoc decision repair module, not a new receiver backbone.

V. Experiments

A. Setup

We evaluate the frozen corrector on top of CURRENTJOINTand compare it with the classical baseline JOINTOMPMPand the intermediate detector-side routes that motivated the final design. The key compared routes are: the base learned receiver, the variance-recalibrated detector variant VX_RECAL_V1, the cleaned mean-residual distillation line, a split-specific observable direct-bit probe, and the final frozen corrector. The main metrics are absolute BER reduction, paired-bootstrap confidence intervals, packet success, and relative goodput gain.

B. From failed residuals to decision-level correction

The corrected same-test-set comparison is the first essential result: the original detector-side mean-residual story does not survive once the evaluation is fixed. Both VX_RECAL_V1and the cleaned distillation model fail to separate meaningfully from CURRENTJOINTon the corrected protocol. This negative result is important because it justifies the paper's shift away from residual regression.

The decision audit then explains what is missing from the residual view. On the corrected held-out cache, the base receiver BER is 6.0352×10^{-2}, the teacher disagreement rate is 5.2051×10^{-2}, and the oracle-beneficial flip rate is 3.1543×10^{-2}. These numbers imply that there is recoverable headroom, but it is concentrated in sparse, local decision errors. A direct bit-level probe confirms that this direction is viable: the split-specific observable corrector yields gains between 6.8359×10^{-4} and 9.2773×10^{-4} on fresh caches.

C. Nominal frozen train-once/test-on-many evaluation

The final method is trained once on one dedicated cache and then frozen. On three untouched nominal caches, the mean BER improves from 6.3553×10^{-2} to 6.2493×10^{-2}, giving an average gain of 1.0590×10^{-3}. The gain standard deviation across the three untouched caches is only 5.5328×10^{-5}, which is small relative to the gain magnitude. This makes the nominal result reproducible enough to support the frozen post-hoc story.

D. Regime transfer and explicit failure boundary

Figure 2 summarizes the regime-level behavior. The same frozen checkpoint remains positive under three nontrivial AFDM shifts:

TABLE I: Method evolution from failed residual correction to the final frozen direct-bit corrector.

Method	Family	Test protocol	Headline outcome
JointOMPMP	Classical iterative baseline	direct evaluation	historical anchor baseline
CurrentJoint	Learned baseline	direct evaluation	base receiver for the final method
vx_recal_v1	Detector calibration	corrected same-test-set evaluation	no meaningful BER change
distill_pure_iter0	Mean-residual distillation	corrected same-test-set evaluation	headline claim collapses after correction
Observable direct-bit corrector	Split-specific probe	split-specific evaluation	gain 6.8359e-04 to 9.2773e-04
Frozen direct-bit corrector	Final method	train-once/test-on-many	nominal gain 1.0590e-03; survives 3 shifts

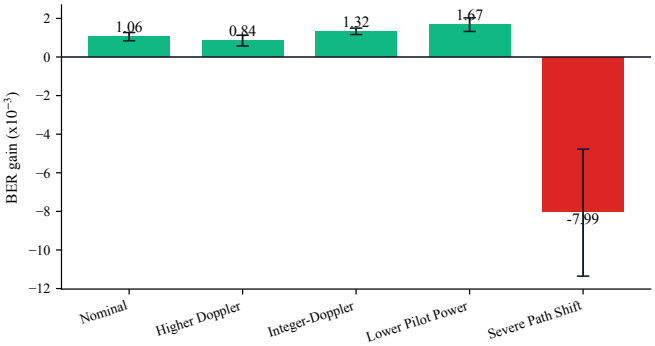


Fig. 2: BER gain of the frozen corrector across evaluation regimes. Error bars show paired-bootstrap 95% confidence intervals. Positive regimes remain above zero, while the severe path-count shift is cleanly negative.

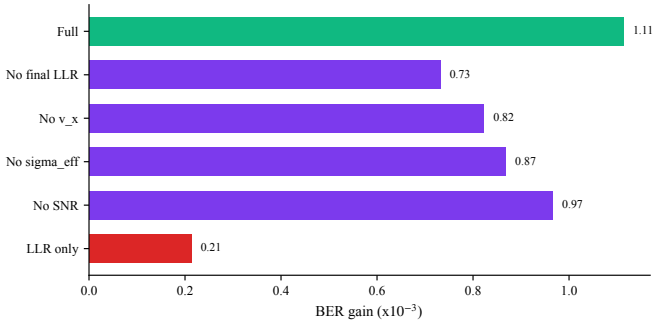


Fig. 3: Feature-group ablation of the frozen corrector. The full observable feature set is strongest, and an LLR-only corrector retains only a small fraction of the full gain.

- higher Doppler severity: mean gain 8.4310×10^{-4},
- integer-Doppler structural shift: mean gain 1.3249×10^{-3} with 95% CI $[1.1621 \times 10^{-3}, 1.4958 \times 10^{-3}]$,
- lower pilot power ($\rho = 0.4$): mean gain 1.6699×10^{-3}.

The paired-bootstrap intervals for the positive regimes all stay above zero. In contrast, a severe path-count shift with $P_{\text{paths}} = 15$ produces a BER regression of -7.9883×10^{-3}, with a confidence interval that stays below zero. This negative regime is not a nuisance result to hide; it is a central part of the paper's scope claim.

E. Feature ablation

The ablation in Figure 3 shows that the method is not simply applying a confidence threshold on the final LLR. The full feature set reaches 1.1133×10^{-3} mean gain, whereas the LLR-only variant falls to 2.1267×10^{-4}. Removing final LLR, v_x, σ_{eff}, or SNR each degrades performance relative to the full model. This supports the interpretation that the corrector uses joint evidence from trajectory evolution and uncertainty, not a single reliability scalar.

F. Packet-level utility

The final reviewer objection was that the BER improvement is small in absolute magnitude. We therefore convert the frozen-corrector predictions into contiguous-packet success and an ARQ-style goodput proxy. Figure 4 shows that the small BER

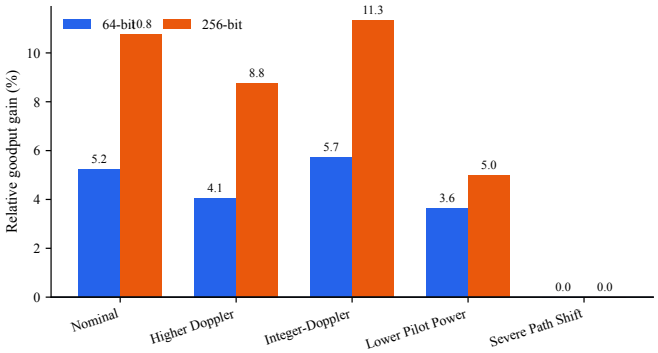


Fig. 4: Relative goodput gains for 64-bit and 256-bit contiguous packets. Small BER gains can amplify into meaningful endpoint gains in the positive regimes, while the severe path shift loses packet-level benefit.

gains are amplified at the packet level. For 64-bit packets, the relative goodput gains range from 3.62% to 5.72% across the positive regimes. For 256-bit packets, the gain ranges from 4.97% to 11.33%. The severe path-count shift remains qualitatively different: it loses reliable benefit at 64 bits and collapses to exactly zero gain at 256 bits.

Overall, the experiments support a scope-bounded claim: a frozen decision-level corrector is a better abstraction than mean-residual distillation in the current AFDM setting, but the benefit is conditional on regime and should not be presented as universal.

TABLE II: Claim and scope matrix used to keep the paper aligned with the validated evidence.

Claim	Key evidence	Scope or limitation
Mean-residual correction is not decision-effective in corrected AFDM evaluation	corrected same-test-set comparison and decision audit	current AFDM joint receiver setting
Frozen decision-level correction yields reproducible BER gains	untouched nominal caches, mean gain 1.0590e-03	train-once/test-on-many inside current simulator family
The gain transfers across 3 regime shifts	positive gains under Doppler, integer-Doppler, and rho=0.4 shifts	10-25 dB, AFDM internal regime shifts
The method has an explicit failure boundary	P_paths=15 regression and negative confidence interval	harsh structural shift breaks the method
Small BER gains can be operationally meaningful	64-bit and 256-bit goodput gains	uncoded contiguous packets with ARQ proxy

VI. Discussion and Limitations

The most important scientific takeaway of this paper is not merely that a small frozen corrector works, but why the earlier route failed. In this AFDM setting, the mismatch is between mean-level regression targets and decision-level utility. Oracle headroom exists, yet most of that headroom is realized through sparse decision-boundary crossings. This is precisely the kind of structure that can evade a residual-distillation objective while still remaining recoverable to a small post-hoc bit-level model.

This perspective also explains why we deliberately do *not* frame the contribution as a new AFDM receiver architecture. The base receiver remains CurrentJoint; the correction module is post-hoc, frozen, and intentionally lightweight. That design improves interpretability and evaluation hygiene, but it also means the absolute BER effect remains modest. The stronger story is therefore diagnostic-plus-method, not architectural replacement.

The paper has three main limitations. First, all positive evidence still lies within the current AFDM simulator family. The regime shifts are meaningful, but they are not out-of-family deployment tests. Second, the system-level endpoint uses uncoded contiguous packets and an ARQ proxy rather than a full coded link stack. Third, the method fails sharply under a severe path-count shift with $P_{\mathrm{paths}} = 15$. We view this explicit boundary as a strength of the paper's scientific honesty, but it is still a real limitation.

These limitations point directly to the next steps: broader transfer beyond the current AFDM simulator family, stronger coded-system endpoints, and simple frozen correctors that may recover a larger fraction of the oracle-beneficial flip set without sacrificing clarity.

VII. Conclusion

We revisited detector-side learning for AFDM joint receivers under corrected evaluation and found that the original mean-residual distillation route does not survive once the evaluation bug is removed. The failure mechanism is informative: the useful headroom is concentrated in sparse decision-boundary crossings rather than smooth residual shifts. Guided by that diagnosis, we introduced a lightweight frozen decision-level corrector that uses only observable receiver trajectory features, improves BER on untouched nominal caches, transfers across three AFDM regime shifts, and yields nontrivial packet-level goodput gains. At the same time, the method keeps a severe path-count shift as an explicit failure boundary. Taken together, these results support a diagnostic-plus-method interpretation of post-hoc AFDM receiver correction: the right intervention is frozen, decision-level, and scope-bounded.

References

[1] A. Bemani, N. Ksairi, and M. Kountouris, "Affine frequency division multiplexing for next generation wireless communications," *IEEE Transactions on Wireless Communications*, vol. 22, no. 11, pp. 8214–8229, 2023. [Online]. Available: http://dx.doi.org/10.1109/TWC.2023.3260906

[2] W. Benzine, A. Bemani, N. Ksairi, and D. Slock, "Affine frequency division multiplexing for communications on sparse time-varying channels," in *GLOBECOM 2023 - 2023 IEEE Global Communications Conference*, 2023, pp. 4921–4926. [Online]. Available: http://dx.doi.org/10.1109/GLOBECOM54140.2023.10437010

[3] H. S. Rou, G. T. F. de Abreu, J. Choi, D. González G., M. Kountouris, Y. L. Guan, and O. Gonsa, "From orthogonal time-frequency space to affine frequency-division multiplexing: A comparative study of next-generation waveforms for integrated sensing and communications in doubly dispersive channels," *IEEE Signal Processing Magazine*, vol. 41, no. 5, pp. 71–86, 2024. [Online]. Available: http://dx.doi.org/10.1109/MSP.2024.3422653

[4] H. Yin, X. Wei, Y. Tang, and K. Yang, "Diagonally reconstructed channel estimation for mimo-afdm with inter-doppler interference in doubly selective channels," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 14 066–14 079, 2024. [Online]. Available: http://dx.doi.org/10.1109/TWC.2024.3408458

[5] H. He, C.-K. Wen, S. Jin, and G. Y. Li, "Model-driven deep learning for mimo detection," *IEEE Transactions on Signal Processing*, vol. 68, pp. 1702–1715, 2020. [Online]. Available: http://dx.doi.org/10.1109/TSP.2020.2976585

[6] N. Shlezinger, R. Fu, and Y. C. Eldar, "Deepsic: Deep soft interference cancellation for multiuser mimo detection," *IEEE Transactions on Wireless Communications*, vol. 20, no. 2, pp. 1349–1362, 2021. [Online]. Available: http://dx.doi.org/10.1109/TWC.2020.3032663

[7] J. Xu, L. Li, L. Zheng, and L. Liu, "Detect to learn: Structure learning with attention and decision feedback for mimo-ofdm receive processing," *IEEE Transactions on Communications*, vol. 72, no. 1, pp. 146–161, 2024. [Online]. Available: http://dx.doi.org/10.1109/TCOMM.2023.3292468